

```
class Cat{
    int eyes=2;
    int tail=1;
    public void jump() {
        System.out.println("Cats jump!");
    }
    public void eatMice() {
        System.out.println("Cats love to eat
mice!");
    }
    public void drinkMilk() {
        System.out.println("Cats drink the
milk while you are away!!");
    }
}
```

```
public class CatTest {

    public static void main(String[] args)
    {
        Cat obj=new Cat();
        System.out.println(obj);
        obj.jump();
        obj.eatMice();
        obj.drinkMilk();
    }
}
```

```
}  
class Dog{  
    //variables  
    int eyes=2;  
    String name="Pluto";  
    float weight=12.4f;  
    int tail=1;  
    String fur="soft";  
    //functions/methods/modules  
    public void jump() {  
        System.out.println("Dogs love to  
jump high");  
    }  
    public void fight() {  
        System.out.println("Dogs fight");  
    }  
    public void bark() {  
        System.out.println("Dogs bark  
loudly");  
    }  
    public void drinkMilk() {  
        System.out.println("Dogs drink  
milk");  
    }  
    public void chaseCat() {  
        System.out.println("Dogs chase the  
cat everytime!!");  
    }  
}
```

```
    }  
}  
public class DogTest {  
  
    public static void main(String[] args)  
    {  
        Dog obj=new Dog();  
        obj.bark();//call the function  
        System.out.println(obj.fur);  
    }  
}
```

```
class Shape{  
    int length=10;  
    int width=20;  
    public void area() {  
        System.out.println("Area");  
    }  
    public void perimeter() {  
        System.out.println("Perimeter");  
    }  
}  
class Square extends Shape{
```

```
}  
public class Test {  
  
    public static void main(String[] args)  
    {  
        Square obj= new Square();  
        obj.area();  
        obj.perimeter();  
        System.out.println("Length  
is"+obj.length);  
        System.out.println("Width  
is"+obj.width);  
    }  
  
}  
  
class Cat{  
    String fur="hardBlack";  
    public void jump() {  
        System.out.println("Cats love to  
jump!!");  
    }  
}  
  
class Kitten extends Cat{  
    String fur="softGrey";  
    public void jump() {
```

```
        System.out.println("Kittens can jump  
but softly!!");  
    }
```

```
}
```

```
public class KittenTest {
```

```
    public static void main(String[] args)  
{  
        Kitten obj=new Kitten();  
        obj.jump();  
        System.out.println(obj.fur);  
    }
```

```
}
```

```
public class Calculator {
```

```
    public void add(int number1,int  
number2) {  
        int sum=number1+number2;  
        System.out.println("Sum of the two  
numbers is "+sum);  
    }
```

```
    public void add(int number1,float  
number2) {  
        float sum=number1+number2;  
        System.out.println("Sum of the two  
numbers is "+sum);  
    }
```

```

    }
    public void add(float number1, float
number2) {
        float sum= number1+number2;
        System.out.println("Sum of the two
numbers is "+sum);
    }
    public void add(float number1, int
number2) {
        float sum=number1+number2;
        System.out.println("Sum of the two
numbers is "+sum);
    }

    public static void main(String[] args)
{
    Calculator obj= new Calculator();
    obj.add(3, 4);
    obj.add(5.6f, 7.8f);
    obj.add(5.6f, 6);
    obj.add(8, 5.8f);

}

}

```

Project:

```
import java.util.Scanner;
class HDFC{
    private int customerId;
    public String customerName;
    float balance=1000000.5f;
    public void withdraw(float
amountWithdraw) {
        balance=balance-amountWithdraw;
        System.out.println("Updated balance
is "+balance);
    }
    public void deposit(float
amountDeposit) {
        balance=balance+amountDeposit;
        System.out.println("Updated balance
is "+balance);
    }
}
class HDFCMumbaiBranch extends HDFC{

}
public class HDFCBankTest {

    public static void main(String[] args)
{
    HDFC obj=new HDFC();
```

```
        System.out.println("Welcome to HDFC  
bank, we are happy to help you!: ");  
        System.out.println("Please press 1  
for withdraw: ");  
        System.out.println("Please press 2  
for deposit: ");  
        System.out.println("Please press 3  
for adding nominee to your account: ");  
        System.out.println("Please press 4  
for adding payee to your account: ");  
        System.out.println("Please press 5  
for exit");  
        Scanner keyBoard=new  
Scanner(System.in);  
        int choice;  
        choice=keyBoard.nextInt();  
        System.out.println("Thank you!. you  
have chosen option "+choice);  
        if(choice==1) {  
            System.out.println("Please enter  
the amount you want to withdraw: ");  
            float  
amountWithdraw=keyBoard.nextFloat();  
            obj.withdraw(amountWithdraw);  
        }  
        else if(choice==2) {
```

```

        System.out.println("Please enter
the amount you want to deposit: ");
        float
amountDeposit=keyBoard.nextFloat();
        obj.deposit(amountDeposit);
    }
    else if(choice ==5) {
        System.out.println("Thank you
for using HDFC bank service, We were happy
to help! :)");
    }

}

}

```

Polymorphism:

1. Overloading:

This is a concept in which we can overload(have a function or an operator) of the exactly same name but different types. Calculator program

2. Overriding:

This is a concept in which the child class if it has same variable and same function overrides the

capabilities/variables/functions of the parent class and uses its own features

Cat-Kitten

Dog-Puppy (Assignment)

Polymorphism: Poly-----morphism

One name and multiple forms

Overloading and overriding are the examples of the polymorphism

Abstraction:

Abstraction is a concept where we show only relevant details to the end user and hide the irrelevant details from the end user.

abstract and interfaces.